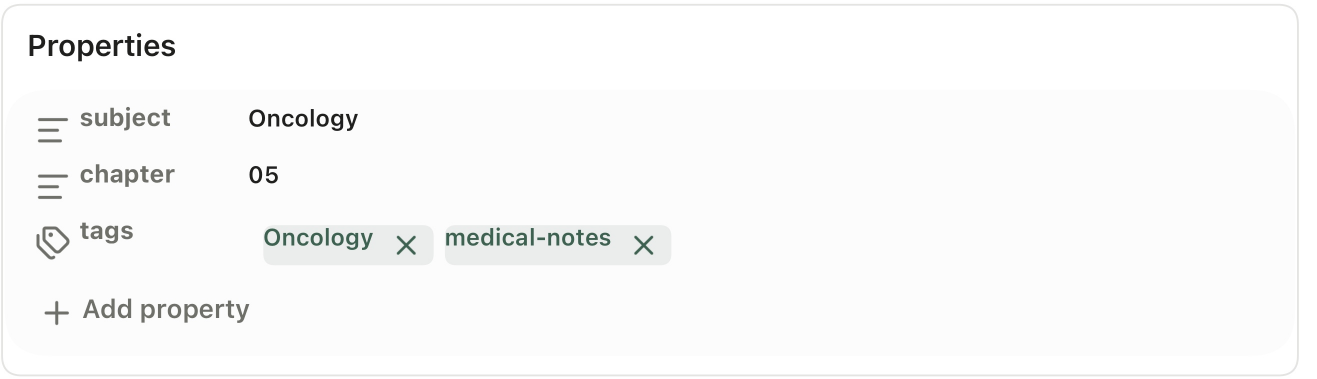
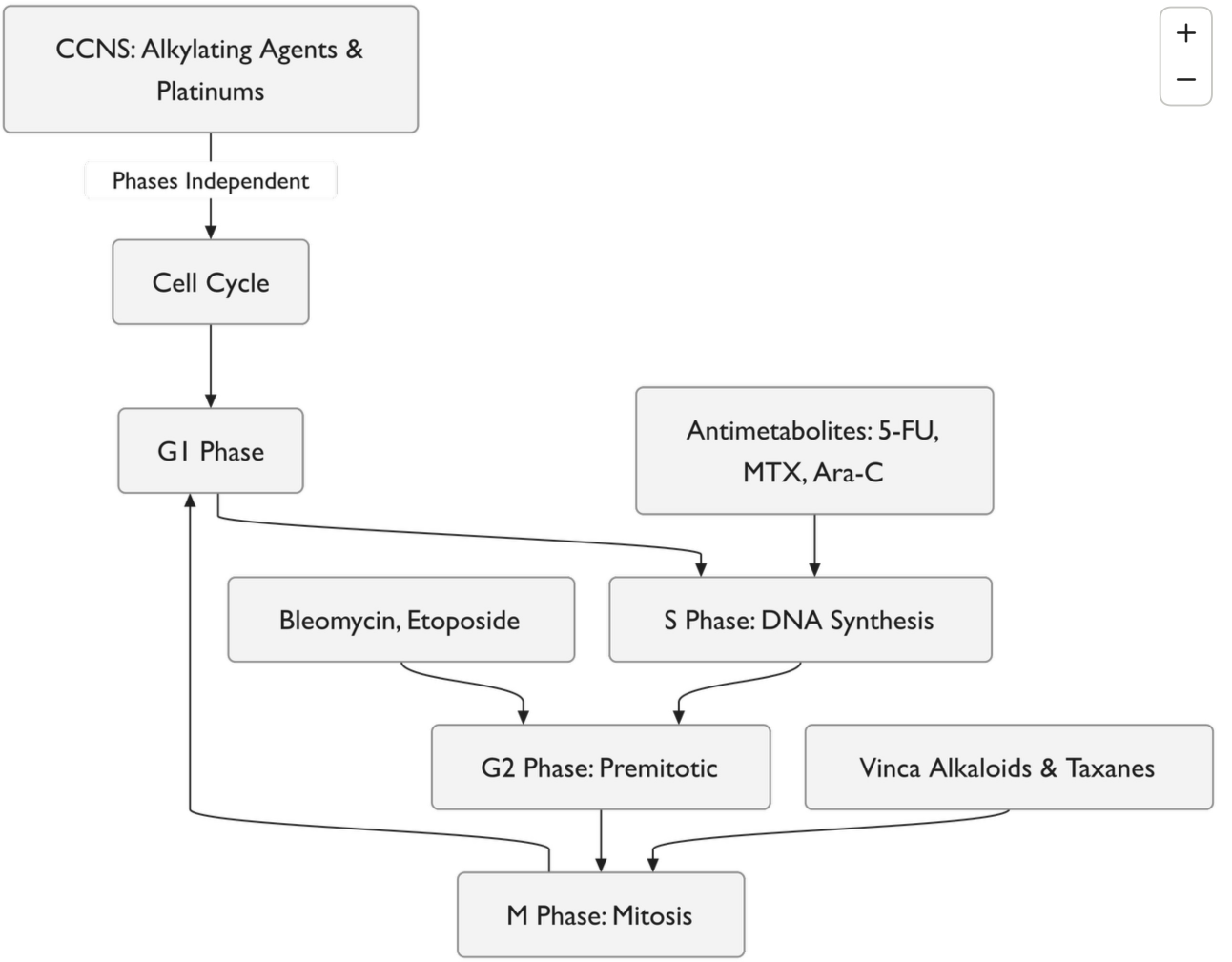
05 Principles Systemic Cancer Therapy



Chapter 5: Principles of Systemic Cancer Therapy

5.1 Cytotoxic Chemotherapy & Pharmacokinetics

Cytotoxic chemotherapy operates primarily by targeting rapidly dividing cells. The fractional killing of cancer cells follows the**Log-Kill Hypothesis**: a given dose of chemotherapy kills a constant*fraction* (percentage)of cells,rather than a constant absolute *number* of cells.



Classification of Cytotoxic Chemotherapeutic Agents

1. Alkylating Agents(Cell Cycle-Nonspecific[CCNS])

- **Mechanism**:Covalently attach alkyl groups to DNA bases(predominantly the $N-7$ position of Guanine),causing DNA cross-linking,strand breaks,and inhibition of DNA replication.
- **Key Drugs**:
    - **Cyclophosphamide & Ifosfamide**: Nitrogen mustard prodrugs activated by hepatic CYP450. Metabolic cleavage yields active phosphoramide

mustard and toxic metabolite **Acrolein**. Acrolein accumulates in the urinary bladder causing **Hemorrhagic Cystitis**. Prevented by aggressive hydration and co-administration of **Mesna** (2-mercaptoethane sulfonate,which binds acrolein in urine).

- **Melphalan**, **Dacarbazine**, **Temozolomide** (crosses blood-brain barrier; used in glioblastoma).

2. Platinum Compounds(CCNS)

- **Mechanism**: Form reactive platinum complexes that bind covalently to $N-7$ positions of guanine and adenine residues, forming intra-strand and inter-strand DNA cross-links.
- **Key Drugs**:
    - **Cisplatin**:High efficacy in testicular,lung,and head & neck cancers. Dose-limiting toxicities: **Nephrotoxicity** (acute tubular necrosis;prevented by pre/post saline hydration)and **Ototoxicity** (irreversible high-frequency sensorineural hearing loss);severe emetogenic potential(requires triple antiemetic prophylaxis).
    - **Carboplatin**:Less nephrotoxic/emetogenic;dose-limiting toxicity is **Myelosuppression** (particularly severe **Thrombocytopenia**). Dosed using the **Calvert Formula**: $\text{Dose (mg)} = \text{Target AUC} \times (\text{GFR} + 25)$.
    - **Oxaliplatin**:Key component of FOLFOX for CRC. Dose-limiting toxicity: **Cold-induced peripheral neuropathy** (acute dysesthesias triggered by cold fluid/air).

3. Antimetabolites(Cell Cycle S-Phase Specific)

- **Mechanism**:Structural analogs of endogenous purines,pyrimidines,or folates that competitively inhibit key metabolic enzymes or incorporate into DNA/RNA.
- **Key Drugs**:
    - **5-Fluorouracil(5-FU)& Capecitabine** (oral prodrug):Inhibits **Thymidylate Synthase(TS)** , preventing dUMP to dTMP conversion. **Leucovorin(Folinic Acid)** is administered concurrently to stabilize the 5-FU-TS-folate ternary complex, **enhancing 5-FU cytotoxicity**. Toxicity: Mucositis,diarrhea,hand-foot syndrome. Severe fatal toxicity occurs in patients with **Dihydropyrimidine Dehydrogenase(DPD)deficiency** .
    - **Methotrexate(MTX)** : Competitively inhibits **Dihydrofolate Reductase (DHFR)**,depleting intracellular tetrahydrofolate(FH_4). **Leucovorin (Folinic Acid) Rescue** is administered 24 hours post-high-dose MTX to bypass DHFR in normal host cells,preventing fatal myelosuppression. Toxicity: Nephrotoxicity(crystalluria),mucositis,hepatotoxicity.
    - **Cytarabine(Ara-C)** :Pyrimidine analog;inhibits DNA Polymerase- α.
    - **Gemcitabine**:Cytidine analog;inhibits Ribonucleotide Reductase.

- **6-Mercaptopurine(6-MP)** : Purine analog. Metabolized by **Xanthine Oxidase**. Co-administration with the xanthine oxidase inhibitor **Allopurinol** blocks 6-MP degradation, leading to fatal toxicity unless the 6-MP dose is reduced by 75%.

4. Antitumor Antibiotics

- **Mechanism**:Intercalate into DNA base pairs,generate free radical oxygen species, and inhibit Topoisomerase II.
- **Key Drugs**:
  - **Anthracyclines(Doxorubicin,Epirubicin)** :Dose-limiting toxicity: **Dilated Cardiomyopathy & Congestive Heart Failure** caused by iron-mediated free radical generation (cumulative lifetime dose limit for Doxorubicin is $450–550\ \mathrm{mg/m^2}$). **Dexrazoxane** (an iron-chelating agent)is administered for cardioprotection in high-dose settings.
  - **Bleomycin**:Glycopeptide antibiotic;binds DNA and generates free radicals causing single/double-strand breaks in G_2/M phase. Dose-limiting toxicity: **Pulmonary Fibrosis** (presents with dry cough,dyspnea,and reduced DLCO; cumulative dose limit $> 400\ \mathrm{units}$). Minimal myelosuppressive activity.

5. Microtubule Inhibitors(Cell Cycle M-Phase Specific)

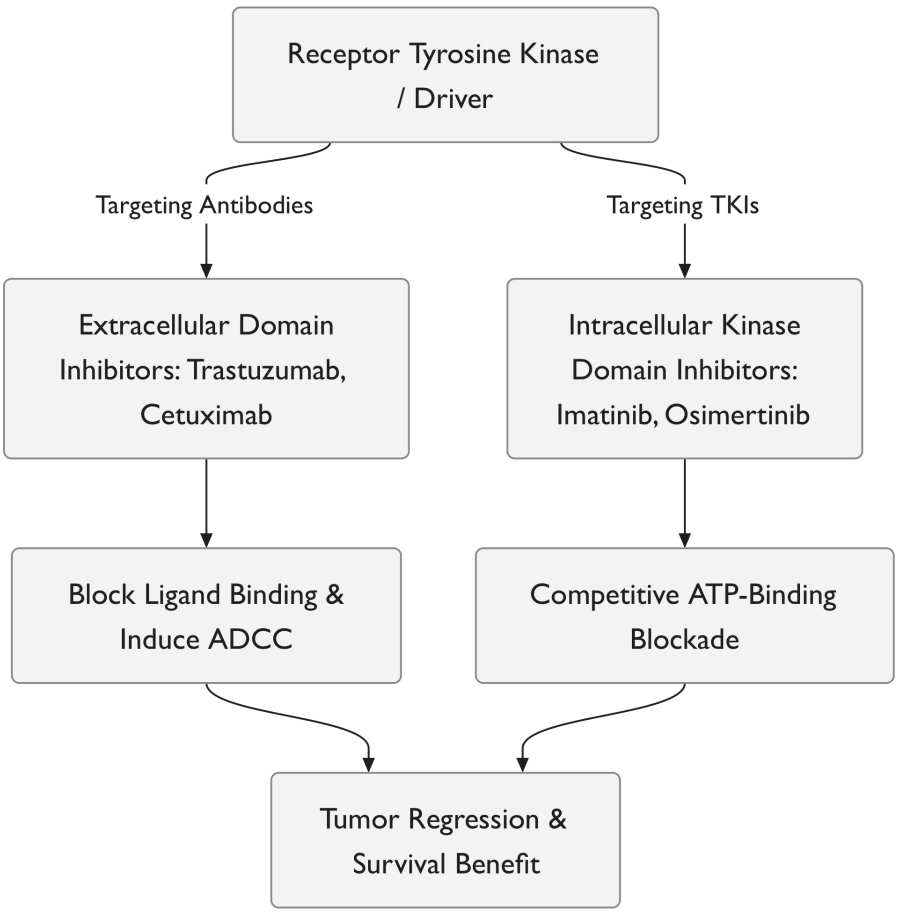
- **Vinca Alkaloids(Vincristine,Vinblastine)** : Bind to β-tubulin and **inhibit tubulin polymerization**,preventing mitotic spindle assembly(M-phase arrest).
  - *Vincristine*:Dose-limiting toxicity: **Peripheral Sensory-Motor Neuropathy** and autonomic neuropathy(paralytic ileus/constipation). Minimal myelosuppression. **Strictly contraindicated for intrathecal administration(fatal neurotoxicity)** .
  - *Vinblastine*:Dose-limiting toxicity:Severe **Bone Marrow Suppression**.
- **Taxanes(Paclitaxel,Docetaxel)** : Bind to β-tubulin and **hyper-stabilize polymerized microtubules**, preventing disassembly and freezing cells in mitosis. Dose-limiting toxicities:Myelosuppression,peripheral neuropathy,hypersensitivity reactions(requires pre-medication with dexamethasone,diphenhydramine,and H_2 blockers).

6. Topoisomerase Inhibitors

- **Topoisomerase I Inhibitors(Irinotecan,Topotecan)** : Irinotecan prevents DNA re-ligation. Causes acute cholinergic syndrome(treated with Atropine)and severe **delayed diarrhea** ("Irinotecan = I-run-to-the-can"). Patients homozygous for $UGT1A1*28$ allele have impaired glucuronidation and elevated severe toxicity risk.
- **Topoisomerase II Inhibitors(Etoposide,Teniposide)** : Etoposide inhibits Topoisomerase II in late S/G_2 phase. Secondary malignancy risk: Treatment-related Therapy-Related AML with $11q23$ (MLL) translocations.

5.2 Targeted Therapy & Small Molecule Kinase Inhibitors

Targeted therapies selectively inhibit specific molecular oncogenic drivers, offering enhanced therapeutic indices over conventional cytotoxic chemotherapy.



Prototypical Small Molecule Tyrosine Kinase Inhibitors(TKIs)

TKI Agent	Primary Molecular Target	Clinical Indications	Key Clinical Considerations / Toxicities
Imatinib	BCR-ABL1, c-KIT(CD117), PDGFR	CML ($t(9;22)$), GIST (*c-KIT* mut), Hypereosinophilic syndrome	First-generation TKI; side effects:fluid retention/ periorbital edema,diarrhea, muscle cramps
Osimertinib	EGFR exon 19 del,L858R,and **T790M resistance**	First-line & resistant *EGFR*-mutated NSCLC	3rd generation irreversible EGFR TKI; spares wild-type EGFR(less rash/diarrhea)

	mutation		
Alectinib	ALK fusion gene rearrangements	First-line *ALK*-rearranged NSCLC	2nd generation ALK TKI; superior CNS penetration over Crizotinib
BRAF/MEK Inhibitors (Dabrafenib + Trametinib)	BRAF V600E / V600K kinase	Metastatic Melanoma, *BRAF*-mutated NSCLC, Anaplastic thyroid CA	Combination BRAF + MEK inhibition delays acquired resistance and reduces secondary skin papillomas
Multi-Kinase Inhibitors (Sunitinib, Sorafenib)	VEGFR-1/2/3, PDGFR,FLT3, RET	RCC,HCC,GIST (post-Imatinib)	Anti-VEGF class toxicities: **Hypertension**, hand-foot skin reaction, proteinuria

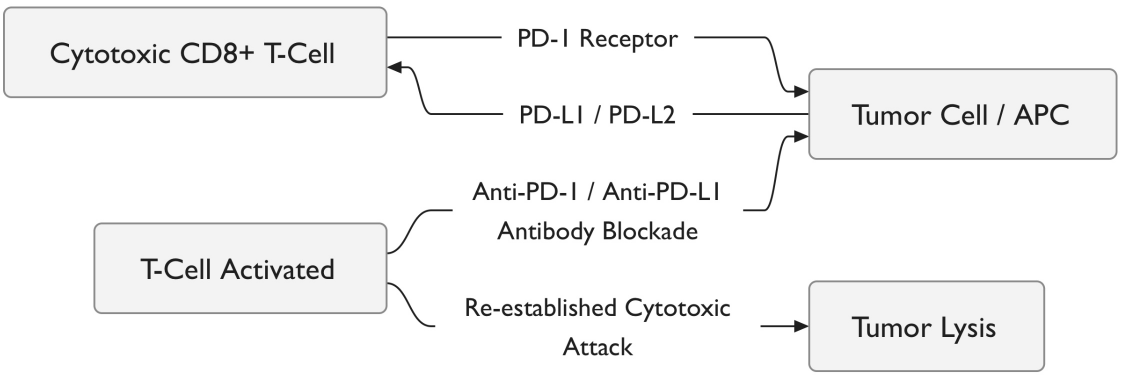
Monoclonal Antibody Targeted Therapies

1. **HER2/neu Inhibitors**:
    - **Trastuzumab**: Humanized IgG1 monoclonal antibody binding domain IV of HER2. Inhibits HER2 homodimerization and induces Antibody-Dependent Cellular Cytotoxicity(ADCC). Used in HER2+ breast and gastric cancers. Dose-limiting toxicity: **Asymptomatic decrease in LVEF / Cardiotoxicity** (unlike anthracyclines,Trastuzumab cardiotoxicity is **not dose-dependent, non-myocyte destructive, and reversible** upon discontinuation).
    - **Pertuzumab**:Binds domain II of HER2,preventing HER2-HER3 heterodimerization. Used in combination with Trastuzumab.
2. **EGFR Inhibitors**:
    - **Cetuximab & Panitumumab**: Bind extracellular domain of wild-type EGFR. Used in metastatic Colorectal Cancer. **Strictly ineffective in tumors with downstream *KRAS*, *NRAS*, or *BRAF* mutations** (mutations bypass upstream EGFR blockade). Toxicity:Acneiform skin rash(severity correlates with tumor response).
3.

3. **Angiogenesis Inhibitors**:
    - **Bevacizumab**: Recombinant humanized IgG1 antibody targeting circulating **VEGF-A**. Prevents VEGFR activation. Toxicities:Hypertension,arterial thromboembolism,impaired wound healing,GI perforation,severe proteinuria.
4. **Antibody-Drug Conjugates(ADCs)**:
    - **Trastuzumab Deruxtecan(T-DXd)** : HER2-directed antibody conjugated to a topoisomerase I inhibitor payload via a cleavable linker. Demonstrates high efficacy in HER2-positive and HER2-Low breast cancers. Toxicity: Interstitial Lung Disease(ILD)/pneumonitis.
    - **Brentuximab Vedotin**: Anti-CD30 antibody conjugated to MMAE (microtubule disruptor);used in Hodgkin Lymphoma and ALCL.

5.3 Cancer Immunotherapy

Cancer immunotherapy aims to harness or re-activate the patient's endogenous immune system to recognize and eliminate neoplastic cells.



1. Immune Checkpoint Inhibitors(ICIs)

- **Mechanisms of Action**:
    - **Anti-PD-1 Antibodies(Pembrolizumab,Nivolumab)** : Block the PD-1 receptor on T cells,preventing engagement by tumor-expressed PD-L1/PD-L2 and reversing T-cell exhaustion.
    - **Anti-PD-L1 Antibodies(Atezolizumab,Durvalumab,Avelumab)** : Directly bind PD-L1 on tumor/immune cells.
    - **Anti-CTLA-4 Antibody(Ipilimumab)** : Blocks CTLA-4 competitively binding CD80/CD86 during early T-cell priming in lymph nodes,expanding memory T-cell pools.
- **Predictive Biomarkers of Response**:
    1. **PD-L1 Expression**:Measured by Tumor Proportion Score(TPS)or Combined Positive Score(CPS).
    2. **Tumor Mutational Burden(TMB-High)** : Defined as $\geq 10\ \text{mutations/Megabase}$. High mutation burden produces abundant neoantigens.
    3. **Mismatch Repair Deficiency(dMMR/MSI-H)** : FDA tissue-agnostic indication for Pembrolizumab.
- **Immune-Related Adverse Events(irAEs)**: Autoimmune-like inflammatory side

effects resulting from systemic T-cell disinhibition:

- *Dermatologic*:Maculopapular rash,pruritus,Vitiligo(common in melanoma).
- *Endocrine*: **Hypophysitis** (common with Ipilimumab;causes pan-hypopituitarism), **Thyroiditis** (hypo/hyperthyroidism),Type 1 Diabetes, Adrenal insufficiency.
- *Gastrointestinal*: Immune-mediated **Colitis** (severe diarrhea;mucosal ulceration).
- *Pulmonary*: Immune-mediated **Pneumonitis** (potential fatal respiratory failure).
- *Management Protocol*:Grade 1(symptomatic support);Grade 2(hold ICI, consider low-dose steroids); **Grade 3–4(permanently discontinue ICI, immediately initiate High-Dose Systemic Corticosteroids [Prednisone** $1–2\ \mathrm{mg/kg/day}$ **or IV Methylprednisolone])**. Steroid-refractory colitis is treated with **Infliximab** (anti-TNF-α).

2. Chimeric Antigen Receptor(CAR)T-Cell Therapy

- **Manufacturing Process**: Autologous peripheral blood T lymphocytes are harvested via leukapheresis, genetically engineered via viral vectors to express a synthetic **Chimeric Antigen Receptor(CAR)**, expanded *in vitro*, and re-infused following lymphodepleting chemotherapy(Fludarabine + Cyclophosphamide).
- **CAR Structure**:Extracellular single-chain variable fragment(scFv)antibody domain(binds target antigen in non-MHC restricted fashion)fused to intracellular signaling domains (**CD3**ζ + co-stimulatory domain **4-1BB** or **CD28**).
- **Clinical Targets & FDA Approvals**:
  - **Anti-CD19 CAR-T** (**Axicabtagene ciloleucel**, **Tisagenlecleucel**): Approved for Relapsed/Refractory Large B-Cell Lymphoma,B-ALL,and Mantle Cell Lymphoma.
  - **Anti-BCMA CAR-T** (**Idecabtagene vicleucel**, **Ciltacabtagene autoleucel**):Approved for Relapsed/Refractory Multiple Myeloma.

- **Life-Threatening CAR-T Toxicities**:
    - **Cytokine Release Syndrome(CRS)**: Massive systemic inflammatory response triggered by explosive CAR-T cell activation and macrophage release of **IL-6**,IL-1,and INF-γ. Symptoms:High fevers,severe hypotension,capillary leak,hypoxia,multi-organ failure. Management: **Tocilizumab** (humanized monoclonal antibody against IL-6 Receptor)$\pm$ Corticosteroids.
    - **Immune Effector Cell-Associated Neurotoxicity Syndrome(ICANS)**: Encephalopathy,dysphasia,tremors,seizures,cerebral edema. Management: **Systemic Corticosteroids(Dexamethasone)** (Tocilizumab does not cross BBB efficiently).

Review & Verification

Subsections Covered Checklist

- [x] ~~5.1 Cytotoxic Chemotherapy & Pharmacokinetics(Log-kill hypothesis,Alkylators Cyclophosphamide/Ifosfamide & Acrolein/Mesna,Platinums Cisplatin/Carboplatin/Oxaliplatin,Antimetabolites 5-FU/Leucovorin/DPD, Methotrexate/DHFR/Leucovorin,Anthracyclines Doxorubicin/Dexrazoxane, Bleomycin,Vincristine vs Vinblastine,Taxanes,Irinotecan/UGT1A1*28).~~
- [x] ~~5.2 Targeted Therapy & Small Molecule Kinase Inhibitors(Imatinib BCR-ABL/c-KIT, Osimertinib T790M,Alectinib,BRAF/MEK,Sunitinib/Sorafenib;Trastuzumab cardiotoxicity,Cetuximab KRAS requirement,Bevacizumab,ADCs T-DXd).~~
- [x] ~~5.3 Cancer Immunotherapy(ICIs Pembrolizumab/Nivolumab/Ipilimumab,biomarkers PD-L1/TMB/MSI-H,irAEs hypophysitis/pneumonitis/colitis management with high-dose steroids/infliximab,CAR-T CD19/BCMA & CRS Tocilizumab/ICANS Dexamethasone).~~

Corrections & Refinements Made

- Clarified that Leucovorin *enhances* 5-FU toxicity/efficacy by stabilizing the TS complex, whereas it *rescues* host cells from Methotrexate by bypassing DHFR.
- Noted that Trastuzumab cardiotoxicity is non-dose-dependent and reversible, contrasting with anthracycline-induced irreversible myocyte destruction.

Source Grounding & Reference Notes

- Grounded in *Robbins Pathology*, *First Aid for the USMLE Step 1(2025)* (Pharmacology section),and current ASCO/NCCN systemic therapy guidelines. All drug mechanisms,toxicities,antidotes,and biomarker requirements match current medical oncology standards.